AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 34 · STUDENT LESSON

Test Diverse Users

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Week 24: rates need the relevant denominators.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Evaluate across varied cases. Try the recall before opening notes.

Your learning target

I can evaluate across varied cases.

I can report failures with their denominators.

I can separate development **evidence** from final checks.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Build the Recommender. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Permissions and Human Review. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Prompt Experiments. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 34: Test Diverse Users

Evaluation must include ordinary, edge, and **underserved** cases.



THINKER MISSION

Run four user profiles; log errors and **fairness concern**s.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

If final cases guide a code change, are they still untouched evaluation cases?

This week's end goal

By the end, I can evaluate varied case types and prioritise an important failure slice.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Test Diverse Users case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use test diverse users to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A fictional chooser passes 5 of 6 development checks but fails its only tie case. An overall **pass rate** can hide that concentrated weakness.

Evaluate ordinary, boundary, invalid, empty and tie cases separately, using requirements rather than duplicated versions of one easy example.

STUDENT LESSON · 3 OF 18

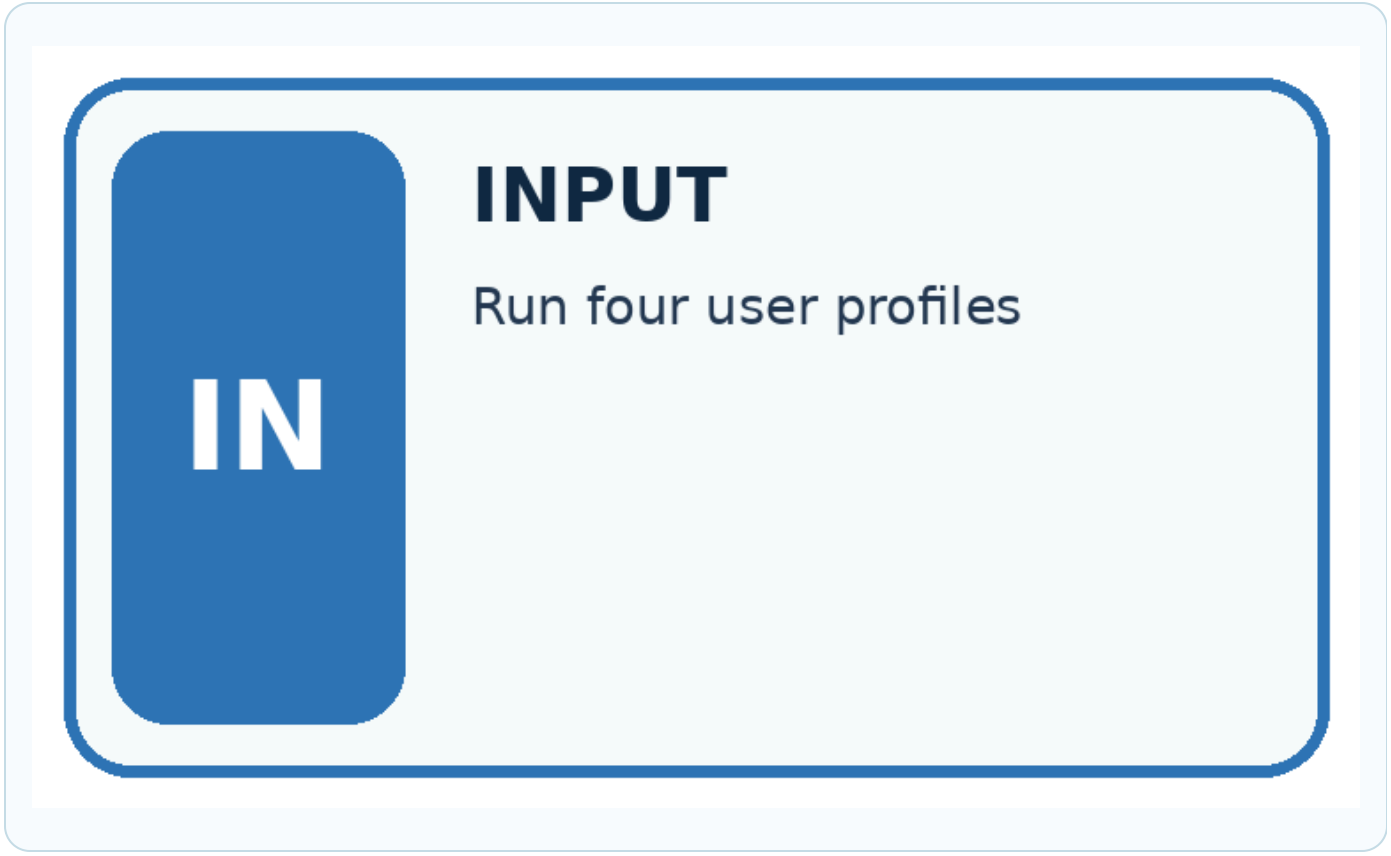
See how it works

> Follow the mechanism for Test Diverse Users. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

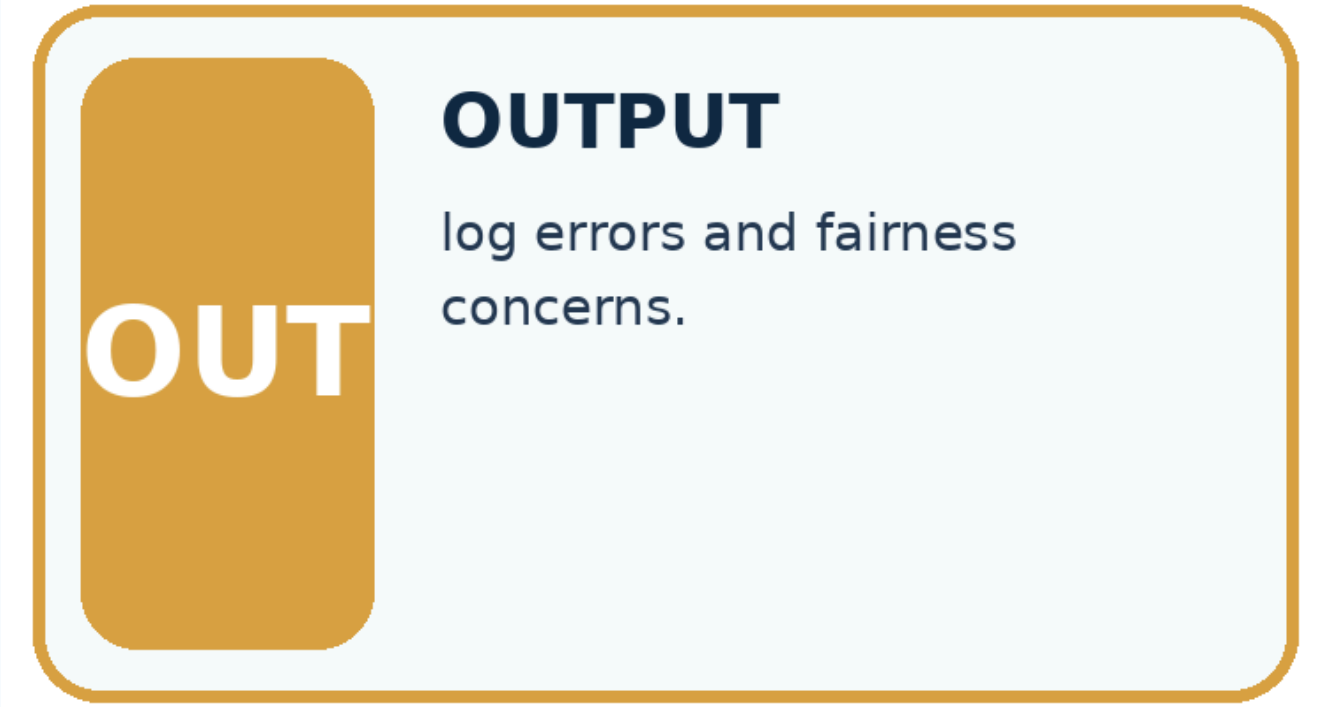
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Run four user profiles

↓


PROCESS
Evaluation must include ordinary, edge, and underserved cases.

↓



Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

```
checks = [{"category": "ordinary", "passed": True},
          {"category": "ordinary", "passed": True},
          {"category": "boundary", "passed": True},
          {"category": "invalid", "passed": True},
          {"category": "empty", "passed": True},
          {"category": "tie", "passed": False}]
passed = 0
for check in checks:
```

```
    if check["passed"]:
        passed = passed + 1
print(passed, len(checks))
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Overall	5 of 6	One failure remains
Tie category	0 of 1	Known weakness
Other categories	Limited cases	No universal guarantee

Read each row from left to right and connect the evidence to the stated rule.

The script reports 5 and 6. About 83.3 percent of these checks pass, but that is not an accuracy estimate for all future use.

The cases were deliberately chosen, their frequencies are artificial, and each category has few cases.

Look inside the overall result

The six development checks include ordinary, boundary, invalid, empty and tie cases. Five pass, but the only tie case fails. Saying five out of six pass is true; saying the tie rule works is false.

Keep both statements visible. A category table helps you choose the next repair. Adding many copies of passing ordinary cases can raise the overall fraction without checking a new condition or fixing the failed requirement.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

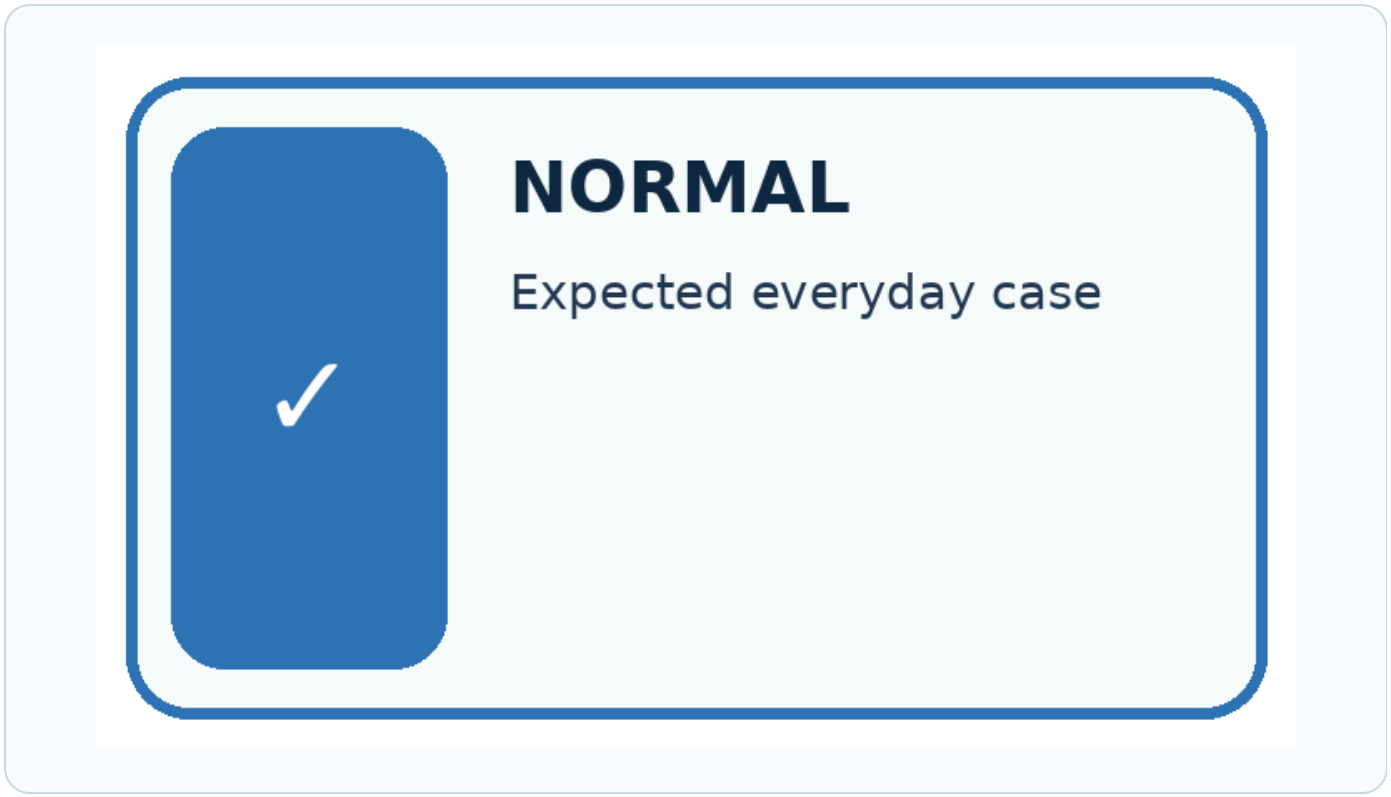
> Compare the supplied cases for Test Diverse Users. Identify the detail that changes the conclusion.

The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

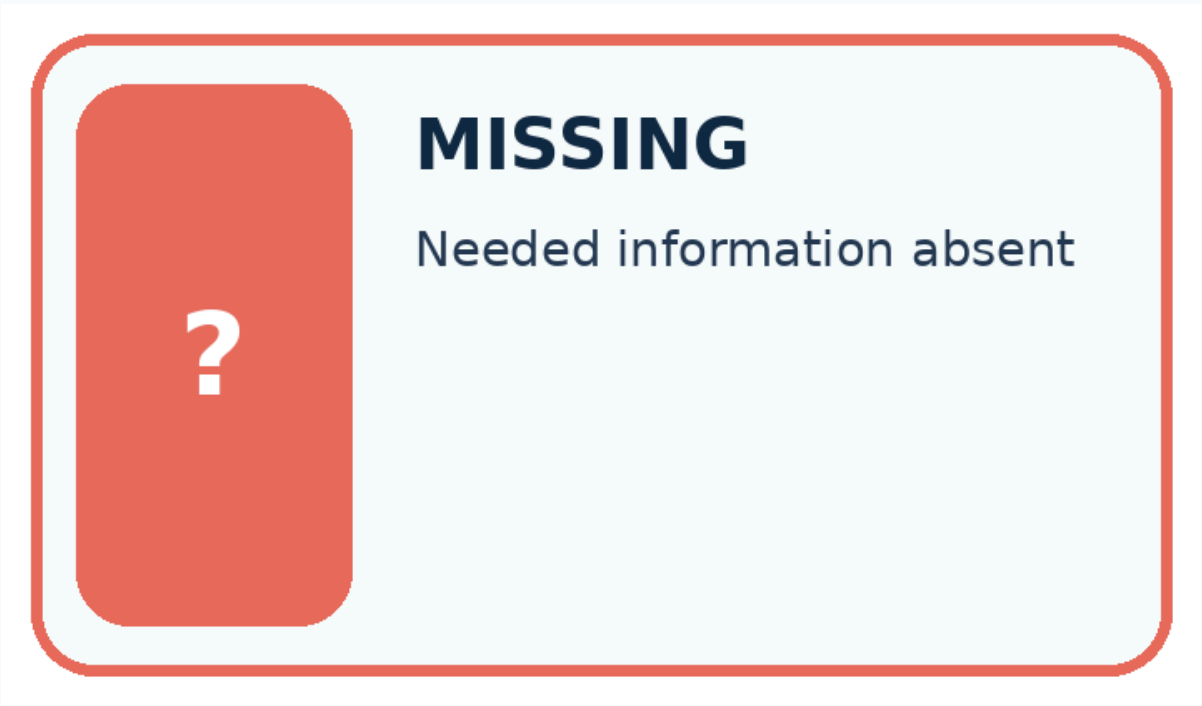
A system is not understood until it survives more than one comfortable example.


NORMAL
Expected everyday case


BOUNDARY
Near a decision edge

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Someone reports only the ordinary slice.

Predicted result

Prediction: The claim becomes 4/4 on ordinary cases, not 100% across all tests.

Why the result follows

Reason: Eight nonordinary cases and their failures have been omitted.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

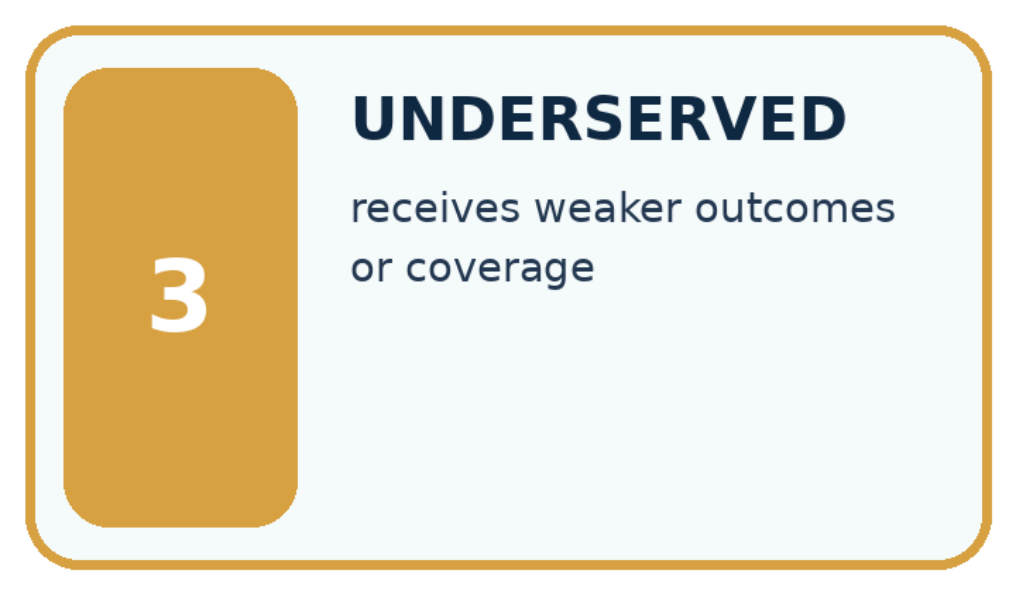
> Use the vocabulary of Test Diverse Users to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
TEST PROFILE
constructed user case for evaluation
2
EDGE CASE
unusual but plausible input





Word	My own example	Not this / boundary
test profile	A description of the kinds of cases and conditions included in an evaluation.	
edge case	An unusual, boundary or difficult input that can reveal a failure hidden by routine examples.	

Word	My own example	Not this / boundary
underserved	Receiving less adequate support or performance than the intended service requires.	
fairness concern	A reason to examine whether a system creates unjustified differences in treatment or outcomes.	

Words used in the closer explanation

Word	Meaning
Evaluation set	Cases used to assess a specified version
Failure category	A type of scenario where an expected result is missed
Pass rate	Passing cases divided by executed cases
Coverage	The variety of requirements and conditions tested

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Test Diverse Users, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 34 • WORKED EVIDENCE

1. Profiles: typical reader; very short-book need; bilingual preference; screen-reader format need.
2. Run the same frozen recommender for all four.
3. Record expected, actual, score evidence, and issue type.
4. Do not average away one group’s repeated failure.

1. Profiles: typical reader; very short-book need; bilingual preference; screen-reader format need.

2. Run the same frozen recommender for all four.

3. Record expected, actual, score **evidence**, and issue type.

4. Do not average away one group’s repeated failure.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

If ten copies of one passing ordinary case are added, what becomes the pass fraction? Does coverage improve?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Test Diverse Users

A fictional recommender is tested on distinct ordinary, equality, invalid-**input** and tie cases.

The facts we will use

Supplied facts

Recorded full passes: ordinary 4/4; equality 1/3; invalid 2/3; ties 1/2.

Supplied facts

Categories are disjoint and every case was run on the same frozen version with the same contract.

AI ACADEMY / AI-2/34

Disjoint evaluation categories

	Full passes	Cases run
Ordinary	4	4
Equality	1	3
Invalid	2	3
Ties	1	2

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/34

Test Diverse Users

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Total count

Passes 4+1+2+1=8; cases 4+3+3+2=12.



2. Overall rate

8/12≈66.7% full passes.



3. Find weak slices

Equality 1/3; invalid 2/3; ties 1/2; ordinary 4/4.



4. Prioritise repair

Inspect failed equality traces and any invalid-input safety consequence, not only the total.

CHANGE ONE THING

Someone reports only the ordinary slice.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Total count	Passes 4+1+2+1=8; cases 4+3+3+2=12.
2. Overall rate	8/12≈66.7% full passes.
3. Find weak slices	Equality 1/3; invalid 2/3; ties 1/2; ordinary 4/4.
4. Prioritise repair	Inspect failed equality traces and any invalid-input safety consequence, not only the total.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Varied tests expose behavior an ordinary demonstration misses. The overall result combines disjoint cases once each; overlapping labels would require unique-case accounting.

Prioritisation also depends on consequences: a malformed **input** that triggers an unauthorised action can be more serious than a harmless tie-order failure. Retain each failed case and its expected **output**.

What this example does not establish

Keep the stated limit

Twelve designed cases describe this test suite, not an unbiased estimate of future usage accuracy.

Keep the stated limit

Read the labelled picture: Can these records establish performance for different user groups?

Keep the stated limit

Change one thing: Someone reports only the ordinary slice. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Test Diverse Users claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Good average performance proves every user is served well. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

For a rule-based chooser, evaluate conformance to its policy separately from whether users find suggestions useful. Neither a pass fraction nor an invented satisfaction score proves educational benefit.

> **You fix the tie bug using the final tie case. What should the next evaluation report say?**

STUDENT LESSON · 8 OF 18

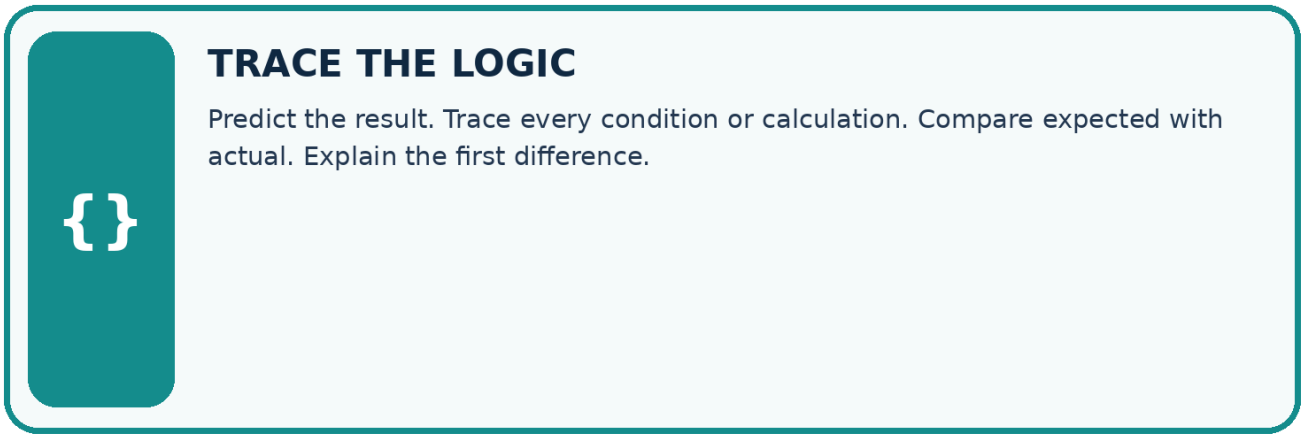
Practise together

> Practise Test Diverse Users with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Hiding the tie failure in an average conceals an unmet requirement. Preserve the failed **input**, expected result, actual result and version.

If you repair using it, classify it as development/regression **evidence** and reserve new final cases.

Practise with less help: Plan coverage

Use workbook W2 for the connected example “Evaluating varied cases”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Evaluating varied cases. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Can these records establish performance for different user groups?

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

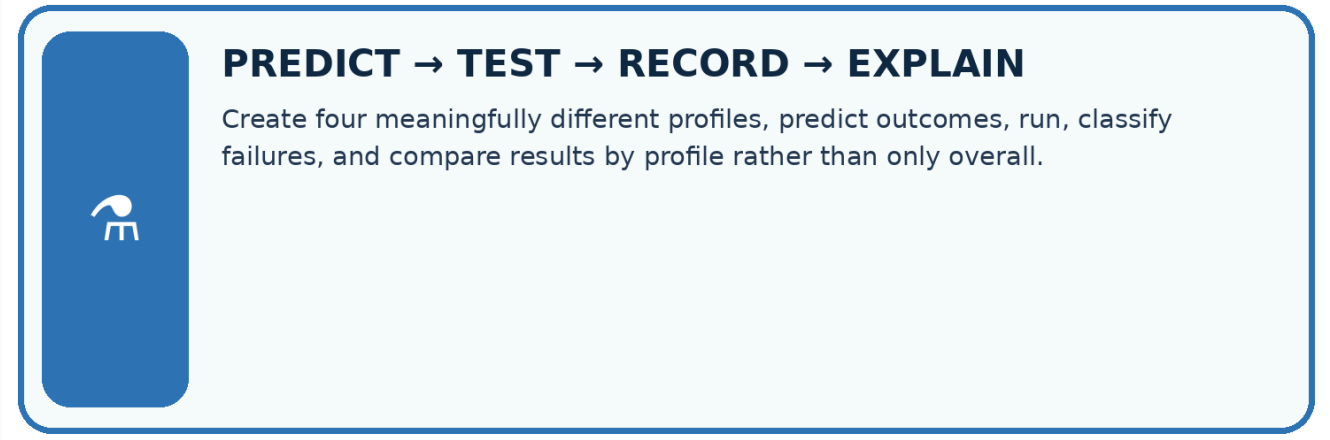
Run the investigation

> Investigate Test Diverse Users. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Create four meaningfully different profiles, predict outcomes, run, classify failures, and compare results by profile rather than only overall.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

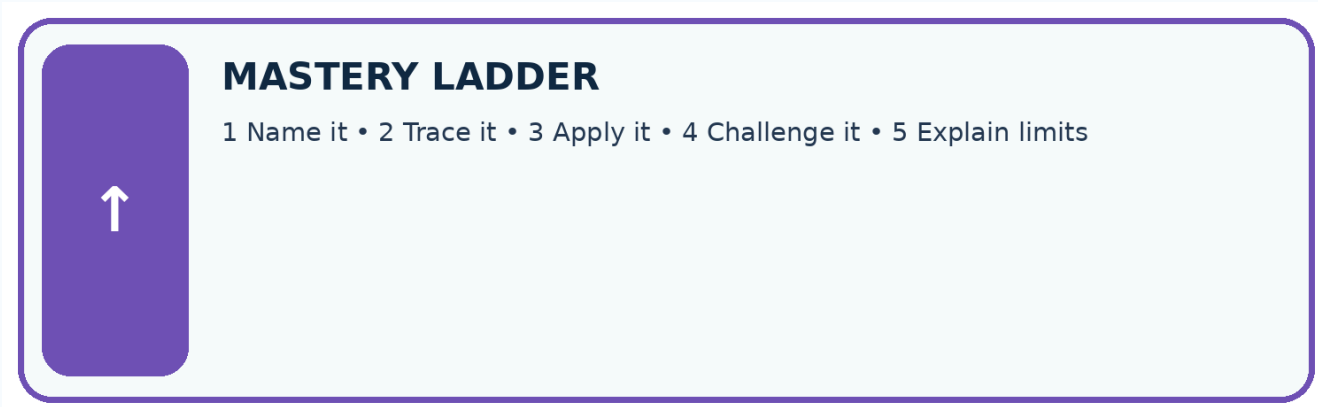
Build your understanding

Move from recall to tracing, diagnosis and transfer in Test Diverse Users. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

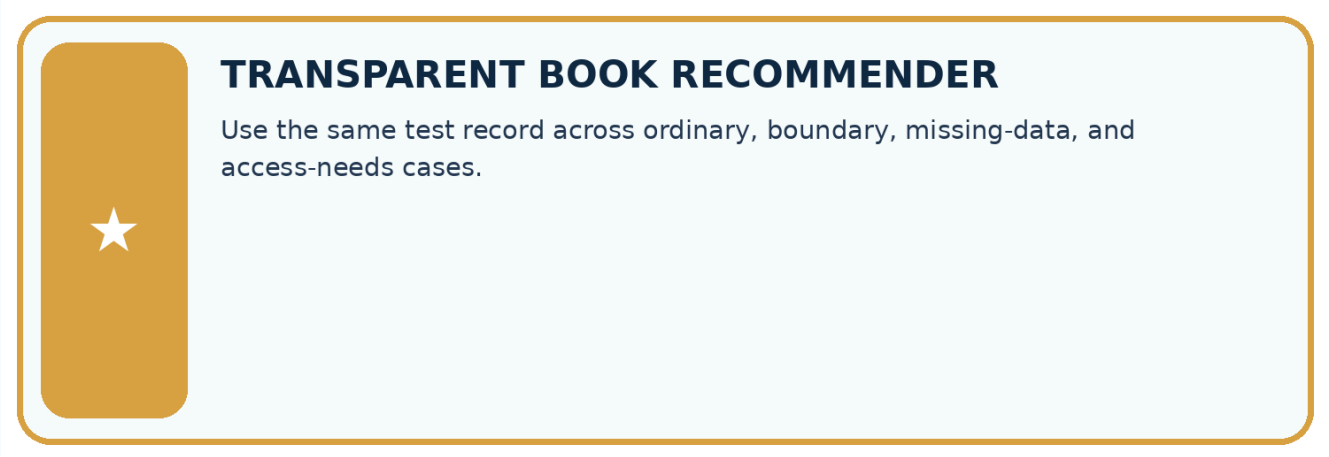
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Use the same test record across ordinary, boundary, missing-data, and access-needs cases.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Build a recommender evaluation matrix with routine, boundary, missing, invalid and tie cases, then add separately planned user-condition **coverage**.

Show what you know

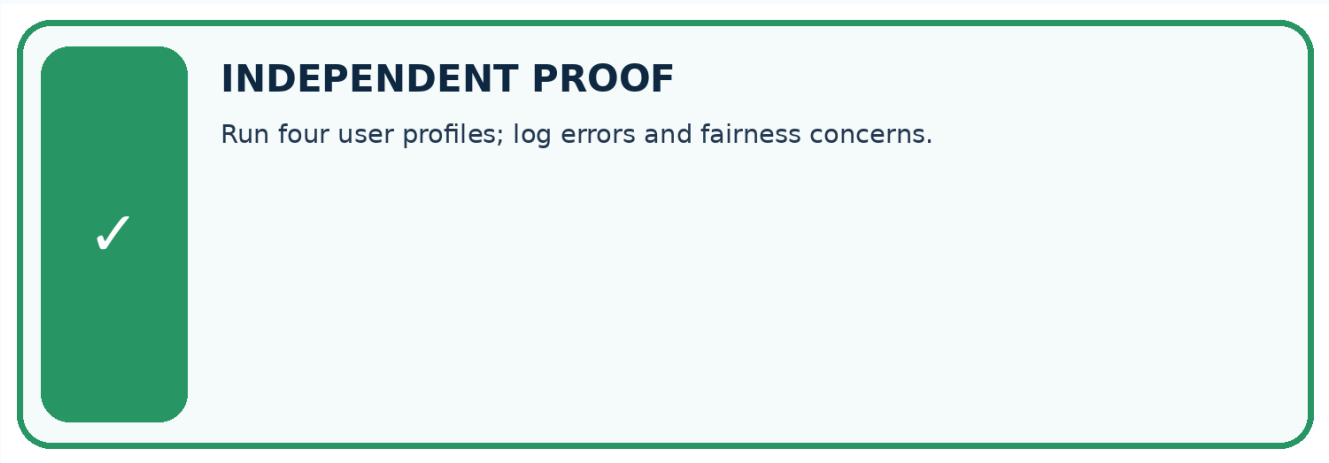
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Run four user profiles; log errors and **fairness concern**s.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, **fairness concern**, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Test Diverse Users case for supported practice. Repeated examples do not become fresh assessments.

Week 34 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A recommender satisfies 5/6 adventure requests and 3/6 poetry requests.

The question

Compare and investigate.

Worked explanation

Rates are 5/6 and 3/6. Inspect catalogue **coverage**, labels and scoring; twelve tests do not establish cause.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Compare success for different interests

Evidence or stage	Value or result
Adventure requests	5/6
Poetry requests	3/6

A recommender satisfies 5/6 adventure requests and 3/6 poetry requests.

Compare and investigate.

Rates are 5/6 and 3/6. Inspect catalogue **coverage**, labels and scoring; twelve tests do not establish cause.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Evaluate independently

Given ordinary 3/3, boundary 1/2, invalid 2/2 and tie 0/1, compute overall results and prioritize issues.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 34 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A recommender serves 6 of 6 easy-book requests but 1 of 4 advanced-book requests. Report both results and identify the next test priority.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Use the same test record across ordinary, boundary, missing-data, and access-needs cases.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

The recommender succeeds on 9/10 mystery requests and 2/5 poetry requests. Compute total and separate rates. A team reports only 11/15. Explain what that hides.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Test Diverse Users, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional recommender is tested on distinct ordinary, equality, invalid-**input** and tie cases.

By the end, I can	How I will show it
Evaluate across varied cases	Explain the deciding step, show a trace or test, and state one limit.
Report failures with their denominators	Explain the deciding step, show a trace or test, and state one limit.
Separate development evidence from final checks	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Test Diverse Users without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Evaluate across varied cases

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional recommender is tested on distinct ordinary, equality, invalid-**input** and tie cases.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Inspect failed equality traces and any invalid-**input** safety consequence, not only the total.

1. Prioritise repair

2. Total count

3. Find weak slices

4. Overall rate

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Prioritise repair Inspect failed equality traces and any invalid-**input** safety consequence, not only the total.

Varied tests expose behavior an ordinary demonstration misses. The overall result combines disjoint cases once each; overlapping labels would require unique-case accounting.

Prioritisation also depends on consequences: a malformed **input** that triggers an unauthorised action can be more serious than a harmless tie-order failure. Retain each failed case and its expected **output**.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Someone reports only the ordinary slice. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The claim becomes 4/4 on ordinary cases, not 100% across all tests.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The claim becomes 4/4 on ordinary cases, not 100% across all tests. Eight nonordinary cases and their failures have been omitted.

Repair a missing building block

Review Build the Recommender (ai-2/33). Goal: Implement a deterministic recommender. Short practice: Explain why B does not win after D is excluded even though it has fewer pages..

Return to this week and explain what you changed.

Review Permissions and Human Review (ai-2/31). Goal: Explain the **mechanism** and terminology of least privilege and tool scopes. Short practice: Which proposal is allowed, and why is an authorised object alone insufficient?.

Return to this week and explain what you changed.

Review Prompt Experiments (ai-2/28). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: Name the case P passes and Q fails, and explain why it matters..

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

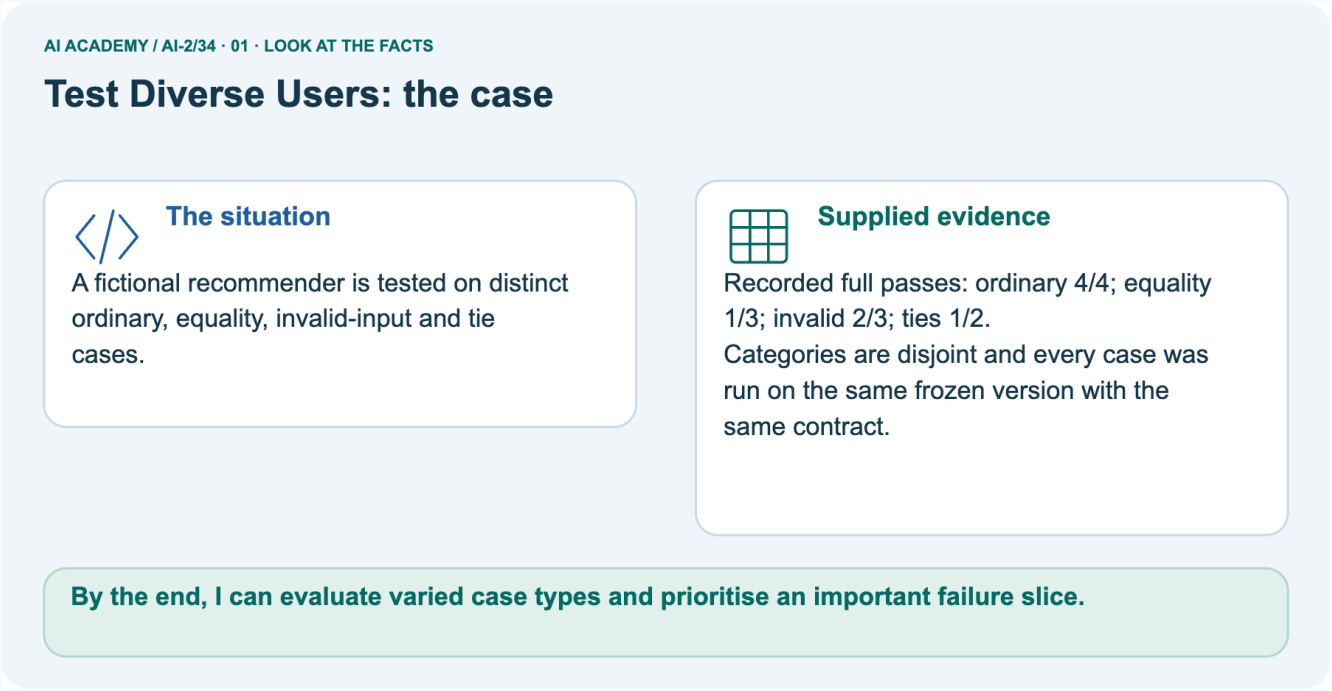
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Test Diverse Users graphic using its labels, deciding evidence and relevant limitation.

01. Test Diverse Users: the case



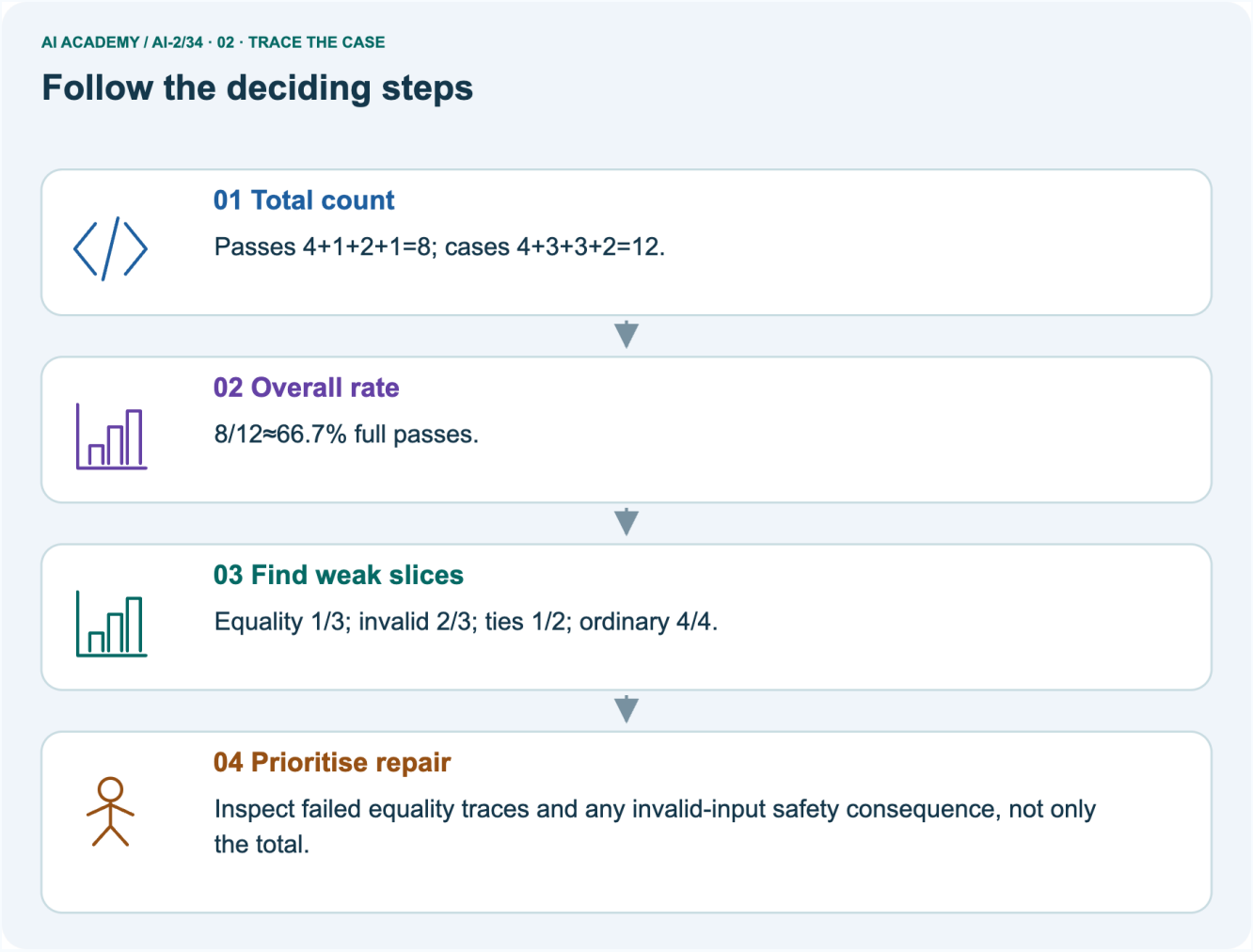
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional recommender is tested on distinct ordinary, equality, invalid-**input** and tie cases.
Supplied facts: Recorded full passes: ordinary 4/4; equality 1/3; invalid 2/3; ties 1/2.

Categories are disjoint and every case was run on the same frozen version with the same contract.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

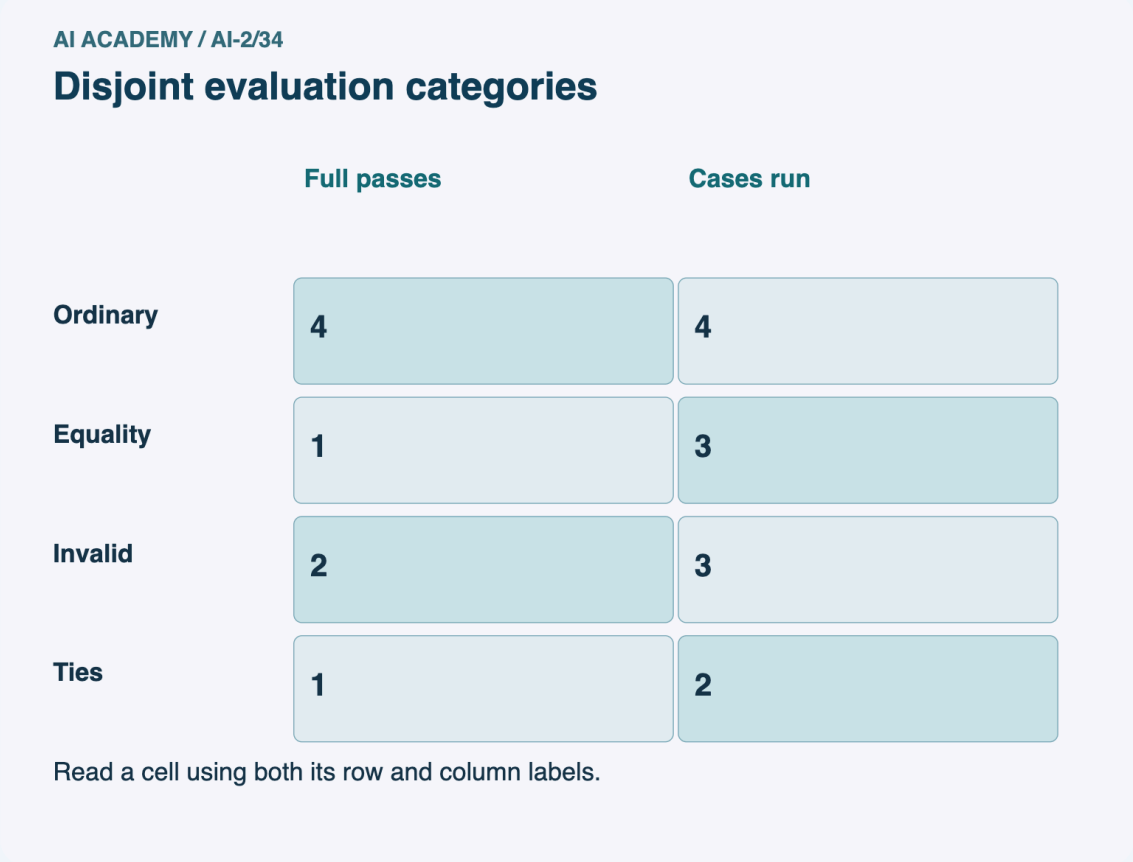


Follow the labelled picture with the matching explanation. Open the vector diagram

Total count: Passes 4+1+2+1=8; cases 4+3+3+2=12. → Overall rate: 8/12≈66.7% full passes. → Find weak slices: Equality 1/3; invalid 2/3; ties 1/2; ordinary 4/4. → Prioritise repair: Inspect failed equality traces and any invalid-**input** safety consequence, not only the total.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Disjoint evaluation categories

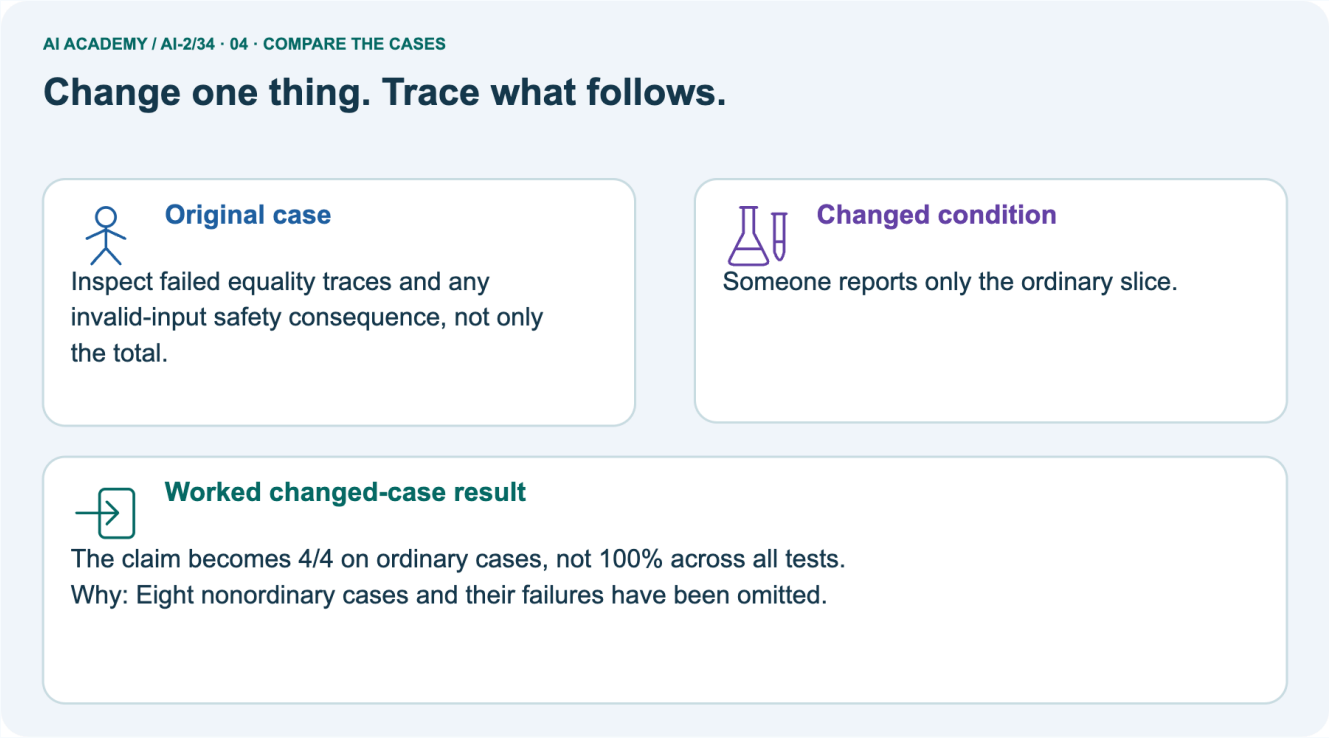


Follow the labelled picture with the matching explanation. Open the vector diagram

Disjoint evaluation categories. Ordinary: Full passes = 4, Cases run = 4. Equality: Full passes = 1, Cases run = 3. Invalid: Full passes = 2, Cases run = 3. Ties: Full passes = 1, Cases run = 2.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



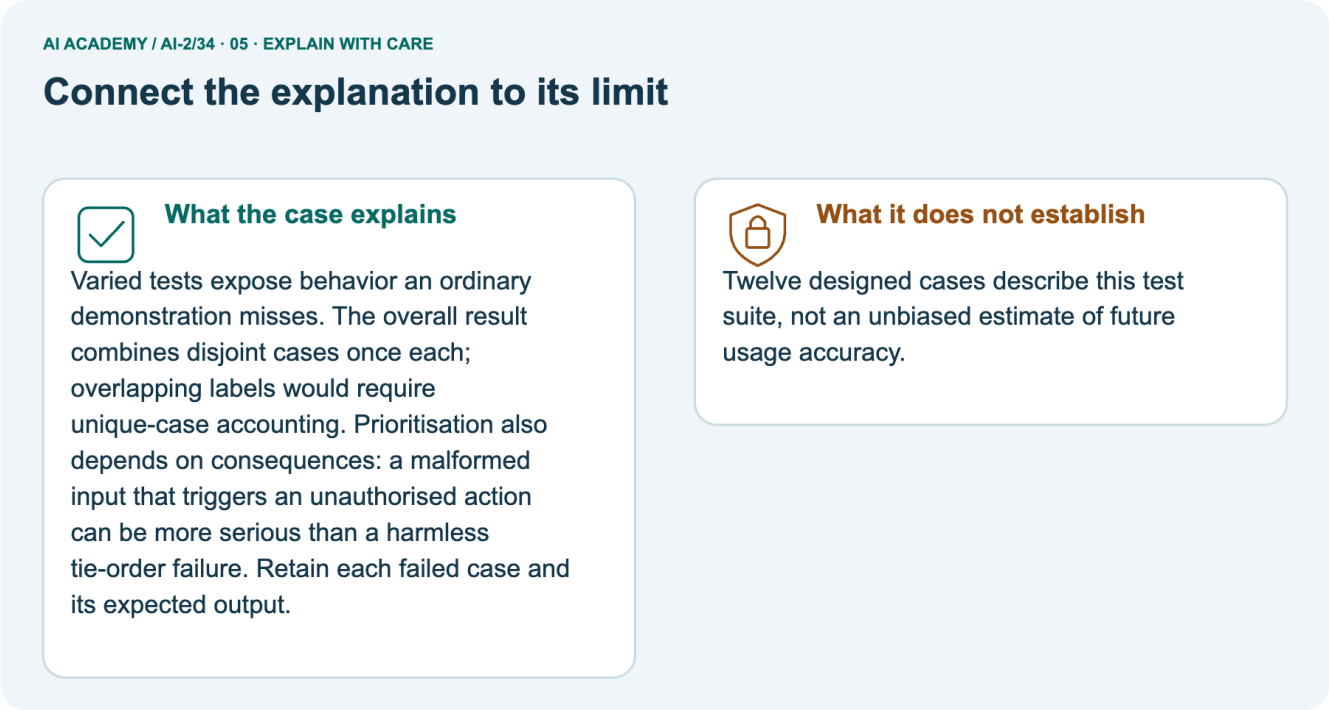
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Inspect failed equality traces and any invalid-**input** safety consequence, not only the total. Changed condition: Someone reports only the ordinary slice.

Expected result: The claim becomes 4/4 on ordinary cases, not 100% across all tests. Reason: Eight nonordinary cases and their failures have been omitted.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

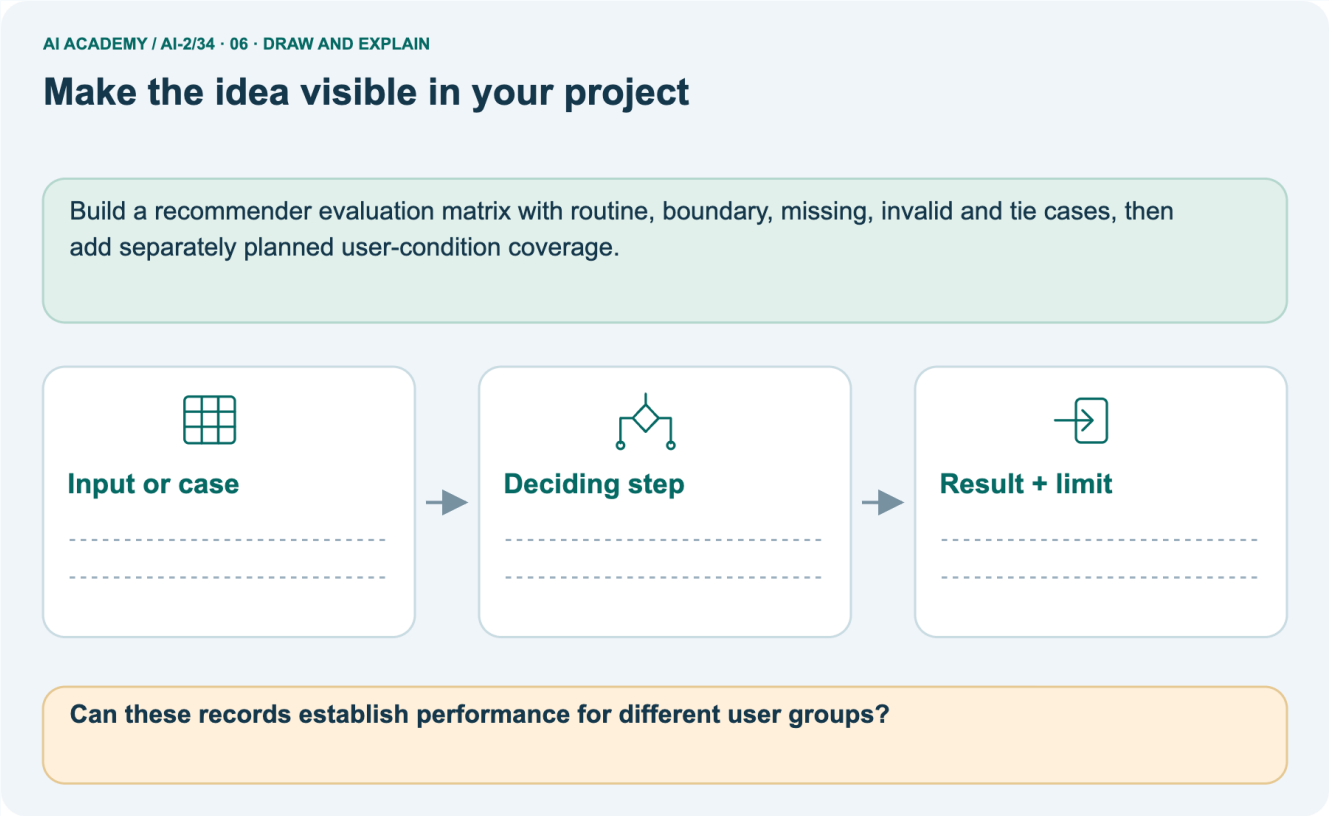
Varied tests expose behavior an ordinary demonstration misses. The overall result combines disjoint cases once each; overlapping labels would require unique-case accounting.

Prioritisation also depends on consequences: a malformed **input** that triggers an unauthorised action can be more serious than a harmless tie-order failure. Retain each failed case and its expected **output**.

Limit: Twelve designed cases describe this test suite, not an unbiased estimate of future usage accuracy.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Build a recommender evaluation matrix with routine, boundary, missing, invalid and tie cases, then add separately planned user-condition **coverage**. Draw your **input**, deciding step, result and limit.

Supported question: Can these records establish performance for different user groups?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.